

ZHAW SCHOOL OF ENGINEERING
INSTITUTE OF MECHATRONIC SYSTEMS (IMS)

Specifications

Betaflight – Offline Controller Tuning

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1 Introduction

First-Person-View (FPV) racing drones require precise and responsive control in order to achieve stable flight behaviour, suppress disturbances such as propeller wash and maintain agility during rapid manoeuvres. The quality of this behaviour strongly depends on the tuning of the flight controller's feedback loops. In particular, the parameters of the angular rate controller determine how quickly and accurately the drone reacts to pilot inputs and external disturbances.

In practice, controller tuning is still predominantly performed manually through iterative trial-and-error procedures. This process requires extensive flight testing and considerable piloting experience, making it both time-consuming and difficult to reproduce. Even though modern flight controller firmware such as Betaflight offers extensive configuration possibilities and detailed flight data logging, efficient and systematic tuning remains a significant challenge.

A previous project introduced a proof-of-concept approach for offline tuning of angular rate controllers using recorded flight data and system identification techniques. Since then, this concept has been further developed beyond the initial proof of concept and forms the basis for the present work.

The objective of this bachelor thesis is to further refine and extend this approach into a modular offline tuning framework for multiple control loops used in Betaflight-based flight controllers. In addition to the already developed angular rate controller tuner, new modules will be introduced for tuning the angle controller, the altitude hold controller, and the position controller. These additions aim to extend the tuning framework beyond the inner rate control loop and provide tools for analysing and tuning higher-level control layers.

The framework will be implemented primarily in MATLAB and Python and will provide tools for analysing flight logs, estimating system dynamics and generating controller parameters without requiring repeated flight tests. The software will be developed as an open-source library and accompanied by comprehensive documentation to ensure accessibility and usability for the FPV community.

By providing a structured, data-driven approach to controller tuning, this project aims to simplify the tuning process, improve flight performance and reduce the reliance on manual trial-and-error methods.

2 Goal / Task Description

The goal of this bachelor thesis is to develop a modular offline tuning framework for multiple control loops used in Betaflight-based flight controllers. The work builds upon an already existing tool for offline tuning of the angular rate controller based on recorded flight data.

Within this thesis, the framework will be extended by adding new modules for tuning the angle controller and the altitude hold controller. If time permits, a position controller tuner will also be implemented. The objective is to enable systematic analysis and tuning of several control layers of the flight controller.

The framework will be implemented primarily in MATLAB and Python. Flight logs recorded with Betaflight will be analysed to estimate the system dynamics of the respective control loops.

To obtain suitable excitation signals for system identification, the firmware of the flight controller will be modified to inject a chirp signal into the input of the angle controller, the altitude hold controller and, if implemented, the position controller. Based on the recorded system responses, controller parameters can then be determined offline.

Both the modified firmware and the developed tuning framework will be extensively tested and validated using flight experiments and recorded flight logs in order to verify the functionality and reliability of the proposed approach.

3 Specification

3.1 Technical Specification

For the implementation of the offline tuning framework, the following technical specifications are defined:

- The framework should be implemented in MATLAB and Python.
- The framework should be modular, allowing for the addition of new tuning modules for different control loops.
- The framework should provide tools for analysing flight logs, estimating system dynamics and generating controller parameters.
- The framework should be developed as an open-source library and accompanied by comprehensive documentation.

3.2 Functional Specification

The functional specifications for the offline tuning framework are as follows:

- The framework should be able to process flight logs recorded with Betaflight firmware.
- The framework should be able to estimate the system dynamics of the angle controller, the altitude hold controller and the position controller based on the recorded flight data.
- The framework should be able to generate controller parameters for the angle controller, the altitude hold controller and the position controller based on the estimated system dynamics.
- The framework should provide a user-friendly interface for configuring and running the tuning process.

3.3 Boundary Conditions

The following boundary conditions are defined for the offline tuning framework:

- The framework should be compatible with Betaflight firmware version
- The framework should be designed for use with FPV racing drones equipped with Betaflight-based flight controllers.

4 Costs

The costs associated with the development of the offline tuning framework are estimated as follows:

- Additional GPS modules for testing the position controller tuner

5 Work Packages

The development tasks of the project are divided among the team members according to their technical focus areas.

- **Work Package 1 – Angle Controller Tuning**

Responsible: Janick Dort

Focus: Control engineering and signal processing

Content: Development of the offline tuning approach for the angle controller. This includes the analysis of flight log data, system identification and signal processing as well as the implementation of the angle controller tuner.

- **Work Package 2 – Altitude Hold Controller**

Responsible: Dario Juriatti

Focus: Software development

Content: Implementation of the altitude hold tuning framework, including firmware modifications for chirp signal injection and development of the required software tools.

- **Work Package 3 – Position Hold Controller**

Responsible: Yuri Bianchi

Focus: Drone tuning and experimental testing

Content: Development of the position hold tuning module and evaluation of the tuning results through practical drone tests and tuning experiments.

- **Work Package 4 – Testing and Documentation**

Responsible: All team members

Content: Joint testing of the modified firmware and the developed tuning framework as well as preparation of the final thesis documentation.

6 Milestones

The development of the offline tuning framework is divided into several milestones. Each milestone represents a development step and defines the expected results.

- **Milestone 1 – Development of the Angle Controller Offline Tuner (MATLAB)**
Content: Implementation of an offline tuning tool for the angle controller using MATLAB. The tool should process Betaflight flight logs and estimate the system dynamics of the angle controller. Acceptance criteria: MATLAB tool successfully analyses flight logs and produces tuning results for the angle controller.
- **Milestone 2 – Chirp Signal Implementation for Altitude Hold**
Content: Modification of the flight controller firmware to inject a chirp excitation signal into the altitude hold control loop. Acceptance criteria: Flight logs contain the excitation signal and the system response required for system identification.
- **Milestone 3 – Altitude Hold Tuning Tool (MATLAB)**
Content: Development of an offline tuning module in MATLAB for the altitude hold controller based on recorded flight data. Acceptance criteria: The MATLAB tool can estimate system dynamics and generate controller parameters for altitude hold.
- **Milestone 4 – Angle Controller Tuner in Python**
Content: Porting or reimplementation of the angle controller tuning tool from MATLAB to Python. Acceptance criteria: Python implementation produces equivalent results to the MATLAB version.
- **Milestone 5 – Chirp Signal Implementation for Position Hold**
Content: Extension of the modified firmware to inject a chirp signal into the position hold controller. Acceptance criteria: Flight logs contain usable excitation data for the position hold controller.
- **Milestone 6 – Position Hold Tuning Tool (MATLAB)**
Content: Development of a MATLAB module for analysing the position hold controller and computing suitable controller parameters. Acceptance criteria: MATLAB tool successfully estimates system dynamics and produces tuning parameters.
- **Milestone 7 – Position Hold Tuning Tool in Python (Optional)**
Content: Implementation of the position hold tuning module in Python based on the MATLAB prototype. Acceptance criteria: Python implementation successfully processes flight logs and generates controller parameters.

7 Signatures

The undersigned accept this specification document as the basis for the bachelor thesis.

Ort, Datum

Betreuer
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Ort, Datum

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